

Implementing the Hawkeye Replacement Policy in ChampSim

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1. Top 3 Learnings / Observations

1. Hawkeye uses reuse behavior to make replacement decisions. The main implementation learning was to separate the policy into three components: OPTgen, the PC-based predictor, and RRIP. OPTgen provides an approximation of the optimal replacement behavior from recent accesses. The predictor learns whether an instruction's cache lines are generally cache-friendly or cache-averse, and RRIP converts this classification into an insertion/eviction priority. This separation made the ChampSim adapter relatively thin and kept the core components independently testable.

2. Associativity strongly affects the benefit of Hawkeye. For the Hmmer benchmark, both policies have substantially lower miss rates as associativity increases. At 4-way associativity, Hawkeye is slightly worse than LRU (34.44% vs. 33.42%). At 8-way, Hawkeye becomes slightly better (26.66% vs. 26.89%), and at 16-way the improvement is larger (25.43% vs. 25.94%). Thus, the benefit of a more informed replacement policy is not uniform across cache organizations.

3. Hawkeye gives a modest but consistent improvement on the evaluated workloads at 16-way. Using the default 2 MB, 16-way LLC, Hawkeye reduces the LLC miss rate relative to LRU by 1.97% for Hmmer, 0.66% for Mcf, and 1.65% for Astar. The improvement is workload-dependent, but all three workloads show a positive reduction in this experiment. This illustrates that replacement policies can provide gains even when cache capacity and line size remain fixed.

2. Required Experiments and Plots

2.1 LLC Miss Rate vs. Associativity

The Hmmer trace (456.hmmer-191B.champsimtrace.xz) was evaluated with a fixed 2 MB LLC and 64-byte cache lines. The three configurations were 4-way/8192 sets, 8-way/4096 sets, and 16-way/2048 sets. Each run used 20M warmup instructions followed by 50M simulation instructions.

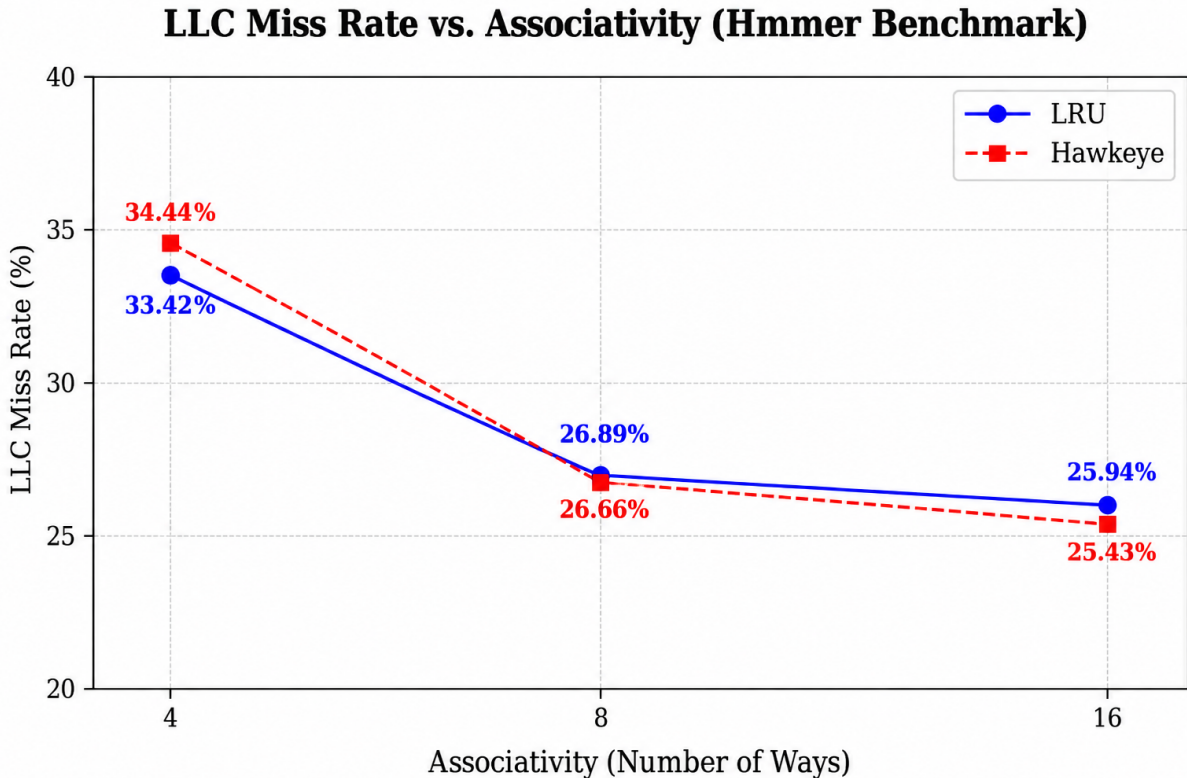


Figure 1: LLC miss rate versus associativity for LRU and Hawkeye on Hmmer.

Associativity	LRU Miss Rate	Hawkeye Miss Rate
4-way	33.42%	34.44%
8-way	26.89%	26.66%
16-way	25.94%	25.43%

2.2 LLC Miss-Rate Reduction over LRU

For the second experiment, the default 2 MB, 16-way LLC configuration was used for Hmmer, Mcf, and Astar. The reduction was calculated as

$$\text{Reduction} = \frac{\text{MissRate}_{LRU} - \text{MissRate}_{Hawkeye}}{\text{MissRate}_{LRU}} \times 100\%.$$

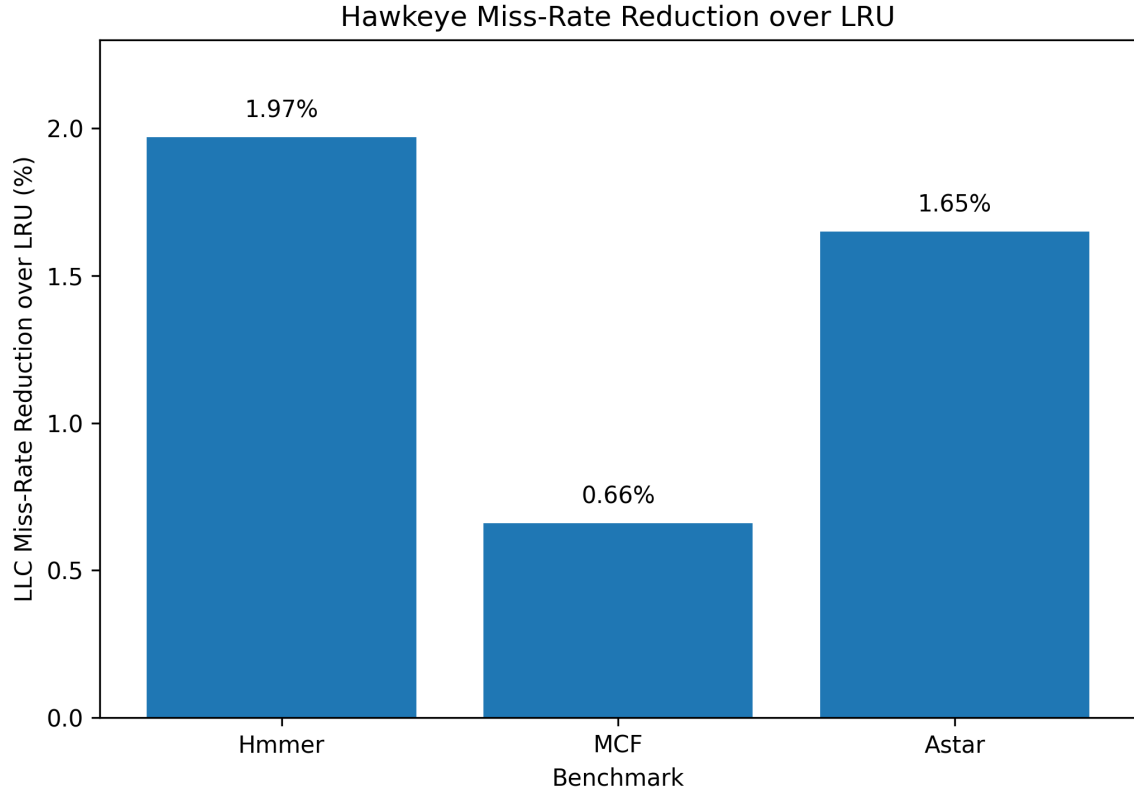


Figure 2: Hawkeye LLC miss-rate reduction relative to LRU at the default 2 MB, 16-way LLC.

Benchmark	Hawkeye Miss-Rate Reduction
456.hmmer-191B	1.97%
429.mcf-22B	0.66%
473.astar-42B	1.65%

Conclusion

The implementation demonstrates the complete Hawkeye replacement flow in ChampSim: recent-reference information is processed by OPTgen, reuse behavior is learned by the PC-indexed predictor, and RRIP performs the final cache insertion and victim selection. The experiments show that Hawkeye’s advantage depends on cache associativity and workload, with the strongest observed Hmmer improvement occurring at 16-way associativity.